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Unit - 2 Quantum Mechanical Model of Atom

Evaluate yourself:

1) Given: accelerated potential = 1 keV

The kinetic energy of the electron =

the energy due to accelerating potential.

$$\frac{1}{2}mv^2 = eV$$

$$mv^2 = 2eV$$

$$m^2v^2 = 2meV \Rightarrow (mv)^2 = 2meV$$

$$\Rightarrow mv = \sqrt{2meV}$$

de Broglie wavelength $\lambda = \frac{h}{mv}$

$$\Rightarrow \lambda = \frac{h}{\sqrt{2meV}}$$

m = mass of the electron = 9.1×10^{-31} kg

h = Planck constant = 6.626×10^{-34} Js

1 eV = 1.6×10^{-19} J

$$\lambda = \frac{6.626\times10^{-34}\,Js}{\sqrt{2\times9.1\times10^{-31}\,kg\times1keV}}$$

$$\lambda = \frac{6.626\times10^{-34}\,JS}{\sqrt{2\times9.1\times10^{-31}\times1\times10^{3}\times1.6\times10^{-19}\,kgJ}}$$

$$\lambda = 3.88\times10^{-11}\,m$$

$$\left[\because \frac{Js}{\sqrt{J\,kg}} = J^{1/2}kg^{-1/2}.\,s\right.$$

$$= (kgm^2s^{-2})^{1/2}.kg^{-1/2}s$$

$$\left. = m\right]$$

2. Given

$\Delta v = 5.7 \times 10^5 ms^{-1}$ Δx = ?

According to Heisenberg's uncertainty principle, $\Delta x \,.\, \Delta p \geq \frac{h}{4\pi}$

$$\frac{h}{4\pi} = \frac{6.626\times10^{-34}}{4\times3.14}\,kgm^2s^{-1}$$

$$\frac{h}{4\pi} = 5.28\times10^{-35}$$

$$\Delta x\,.\,\Delta p \geq 5.28\times10^{-35}$$

$$\Delta x\,.\,m.\Delta v \geq 5.28\times10^{-35}$$

$$\Rightarrow \Delta x \geq \frac{5.28\times10^{-35}\,kgm^2s^{-1}}{9.1\times10^{-31}\,kg\times5.7\times10^{5}\,ms^{-1}}$$

$$\Rightarrow \Delta x \geq 1.017\times10^{-10}\,m$$